

GMV Anomaly Intelligence Report

Olist Brazilian E-Commerce · Jan 2017 – Aug 2018

Generated June 08, 2026

12	11	1	5
Total Anomalies	Spikes ▲	Drops ▼	High Severity

Methodology. Anomalies are detected using a rolling 28-day median + Median Absolute Deviation (MAD) on the detrended daily GMV residual. A day is flagged when $|\text{robust z-score}| \geq 3.5$, where $z = 0.6745 \times \text{residual} / \text{MAD}$. Segment attribution decomposes each flagged day's total delta into per-segment dollar contributions versus that segment's own 28-day baseline. **Three synthetic shocks** were injected into the dataset (2017-05-16, 2017-07-21, 2018-06-30) to validate the detection pipeline — these are disclosed and identifiable by their extreme segment-level magnitudes. All narratives are LLM-generated (Claude Sonnet) from pre-computed facts only; no numbers are invented.

Anomaly Index

Date	Direction	Severity	Z-Score	% vs Baseline	Top Driver	Decomposed By
2017-04-18	▲ Spike	Medium	+3.84	+118.5%	Small Appliances	Category
2017-05-16	▲ Spike	High	+8.34	+150.0%	Sports Leisure	Category
2017-07-21	▲ Spike	Medium	+4.28	+125.9%	Bed Bath Table	Category
2017-11-20	▲ Spike	High	+6.43	+100.1%	Southeast	Region
2017-11-21	▲ Spike	Medium	+3.58	+51.8%	Southeast	Region
2017-11-23	▲ Spike	High	+5.03	+90.3%	Southeast	Region
2017-11-24	▲ Spike	High	+29.79	+558.0%	Southeast	Region
2017-11-25	▲ Spike	High	+8.35	+160.6%	Southeast	Region
2017-11-26	▲ Spike	Medium	+4.28	+90.7%	Southeast	Region
2017-11-27	▲ Spike	Medium	+4.21	+99.3%	Southeast	Region
2017-11-28	▲ Spike	Medium	+3.98	+96.1%	Southeast	Region
2018-06-30	▼ Drop	Medium	-3.82	-52.1%	Southeast	Region

▲ 2017-04-18

SPIKE

MEDIUM SEVERITY

R\$ 21,287

Actual GMV

R\$ 9,744

28-Day Baseline

+118.5%

vs Baseline

+3.84

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by CATEGORY

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Small Appliances	+9,403	+85.7%	+828.5%	▲
Bed Bath Table	+1,007	+9.2%	+137.5%	▲
Other	+947	+8.6%	+53.1%	▲
Health Beauty	-588	-5.4%	-90.0%	▼
Perfumery	+432	+3.9%	+63.3%	▲

WHAT HAPPENED

On 2017-04-18, GMV spiked to 21,287 vs. a baseline of 9,744 (+118.5%), a medium-severity anomaly (z=3.84).

WHY

Small_appliances drove 85.7% of the total move, surging +828.5% above its own baseline — an outsized, category-specific event. Bed_bath_table contributed an additional 9.2% of the move (+137.5% vs. baseline), with remaining categories adding modest upward pressure. Health_beauty partially offset the spike, declining 90.0% vs. baseline and subtracting 5.4% from the total move. No historical precedent is on record.

RECOMMENDED ACTION

Investigate small_appliances immediately: identify whether a flash sale, mis-pricing, bulk/fraudulent order, or catalog error drove the +828.5% surge on this specific date.

▲ 2017-05-16

SPIKE

HIGH SEVERITY

R\$ 34,295

Actual GMV

R\$ 13,720

28-Day Baseline

+150.0%

vs Baseline

+8.34

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by CATEGORY

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Sports Leisure	+16,545	+87.8%	+1612.4%	▲
Watches Gifts	+1,629	+8.6%	+175.1%	▲
Other	+1,052	+5.6%	+44.6%	▲
Small Appliances	-998	-5.3%	-84.7%	▼
Housewares	+762	+4.0%	+138.2%	▲

WHAT HAPPENED

On 2017-05-16, GMV spiked +150.0% above baseline (actual: 34,295 vs. 13,720; z=8.34), marking a high-severity anomaly.

WHY

The move was overwhelmingly driven by sports_leisure, which surged +1,612.4% vs. baseline and contributed 87.8% of the total uplift. watches_gifts added a further 8.6% of the move (+175.1% vs. baseline). Notably, a comparable spike occurred exactly 28 days prior on 2017-04-18, suggesting a recurring monthly pattern. small_appliances partially offset gains, declining -84.7% vs. baseline (-5.3% of the move).

RECOMMENDED ACTION

Immediately audit sports_leisure orders on 2017-05-16 — check for a promotional event, bulk/fraudulent orders, or data pipeline duplication that could explain the +1,612.4% surge.

■ A spike of similar severity occurred on 2017-04-18 (28 days prior), z=+3.84.

▲ 2017-07-21

SPIKE

MEDIUM SEVERITY

R\$ 35,872

Actual GMV

R\$ 15,877

28-Day Baseline

+125.9%

vs Baseline

+4.28

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by CATEGORY

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Bed Bath Table	+24,173	+118.4%	+1224.2%	▲
Computers Accessories	-936	-4.6%	-69.4%	▼
Watches Gifts	+766	+3.7%	+73.8%	▲
Other	-622	-3.0%	-32.0%	▼
Furniture Decor	-608	-3.0%	-74.9%	▼

WHAT HAPPENED

On 2017-07-21, GMV spiked to 35,872 versus a baseline of 15,877 (+125.9%, z=4.28), marking a medium-severity anomaly—exactly 66 days after the prior spike on 2017-05-16.

WHY

bed_bath_table drove virtually the entire move, contributing +118.4% of the total shift with the segment itself surging +1,224.2% above baseline. watches_gifts added a modest +3.7% contribution (+73.8% vs. baseline). Offsetting factors included computers_accessories (-4.6% contribution) and furniture_decor (-3.0% contribution), both declining sharply. Notably, the previous anomaly on 2017-05-16 showed identical 66-day spacing, suggesting a possible recurring promotional cycle.

RECOMMENDED ACTION

Investigate bed_bath_table immediately—identify whether a targeted promotion, bulk order, or data pipeline event caused the +1,224.2% surge, and cross-reference with 2017-05-16 order records to confirm whether the same mechanism triggered both anomalies.

■ A spike of similar severity occurred on 2017-05-16 (66 days prior), z=+8.34.

▲ 2017-11-20

SPIKE

HIGH SEVERITY

R\$ 43,921

Actual GMV

R\$ 21,952

28-Day Baseline

+100.1%

vs Baseline

+6.43

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+12,964	+58.7%	+95.5%	▲
Northeast	+7,068	+32.0%	+264.7%	▲
North	+1,114	+5.0%	+170.0%	▲
Center-West	+875	+4.0%	+53.6%	▲
South	+46	+0.2%	+1.4%	▲

WHAT HAPPENED

On 2017-11-20, GMV spiked to 43,921 — a +100.1% increase vs. the 21,952 baseline (z=6.43), the first anomaly in 122 days.

WHY

Southeast drove 58.7% of the total move (+95.5% vs. baseline), followed by Northeast contributing 32.0% (a dramatic +264.7% surge). North and Center-West added a combined 9.0% of the move. Notably, a comparably severe spike occurred exactly 122 days prior on 2017-07-21 (z=4.28), suggesting a potential recurring seasonal or promotional pattern.

RECOMMENDED ACTION

Prioritize investigation of the Northeast segment first — its +264.7% surge is disproportionate to its share of the move, indicating a localized event (flash sale, campaign, or data anomaly) requiring immediate verification.

■ A spike of similar severity occurred on 2017-07-21 (122 days prior), z=+4.28.

▲ 2017-11-21

SPIKE

MEDIUM SEVERITY

R\$ 33,731

Actual GMV

R\$ 22,220

28-Day Baseline

+51.8%

vs Baseline

+3.58

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+6,673	+60.2%	+47.8%	▲
South	+4,028	+36.3%	+120.3%	▲
North	+363	+3.3%	+51.6%	▲
Northeast	+125	+1.1%	+4.2%	▲
Center-West	-99	-0.9%	-5.9%	▼

WHAT HAPPENED

On 2017-11-21, GMV spiked to 33,731 versus a baseline of 22,220 (+51.8%, z=3.58), marking a medium-severity anomaly.

WHY

Southeast drove 60.2% of the total move (+47.8% vs baseline), followed by South contributing 36.3% (itself up +120.3% vs baseline), together accounting for nearly all of the lift. North added a minor 3.3% share. Notably, a higher-severity spike (z=6.43) occurred the prior day (2017-11-20), suggesting a sustained multi-day demand event across these regions. Center-West was the sole drag, offsetting -0.9% of the move.

RECOMMENDED ACTION

Investigate the South region first — its +120.3% segment-level surge is disproportionate and warrants checking for promotional campaigns, fulfillment anomalies, or data integrity issues specific to that region on both November 20–21.

■ A spike of similar severity occurred on 2017-11-20 (1 days prior), z=+6.43.

▲ 2017-11-23

SPIKE

HIGH SEVERITY

R\$ 43,564

Actual GMV

R\$ 22,886

28-Day Baseline

+90.3%

vs Baseline

+5.03

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+13,386	+66.1%	+92.8%	▲
South	+3,579	+17.7%	+102.6%	▲
Northeast	+2,606	+12.9%	+86.4%	▲
North	+455	+2.2%	+67.3%	▲
Center-West	+225	+1.1%	+13.1%	▲

WHAT HAPPENED

On 2017-11-23, GMV spiked to 43,564 — 90.3% above the baseline of 22,886 ($z=5.03$), marking a high-severity anomaly.

WHY

Southeast drove 66.1% of the total move (+92.8% vs baseline), followed by South at 17.7% (+102.6%) and Northeast at 12.9% (+86.4%). This spike follows a similar high-severity event two days prior on 2017-11-21 ($z=3.58$), suggesting a sustained multi-day demand surge across all regions, consistent with a major promotional event.

RECOMMENDED ACTION

Prioritize the Southeast segment for immediate review — confirm whether campaign spend, promotional mechanics, or fulfillment capacity align with its outsized 66.1% contribution to determine if the surge is sustainable or demand has been pulled forward.

■ A spike of similar severity occurred on 2017-11-21 (2 days prior), $z=+3.58$.

▲ 2017-11-24

SPIKE

HIGH SEVERITY

R\$ 151,778

Actual GMV

R\$ 23,066

28-Day Baseline

+558.0%

vs Baseline

+29.79

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+91,983	+72.0%	+614.2%	▲
South	+17,204	+13.5%	+472.5%	▲
Center-West	+9,249	+7.2%	+526.7%	▲
Northeast	+8,842	+6.9%	+294.8%	▲
North	+409	+0.3%	+57.0%	▲

WHAT HAPPENED

On 2017-11-24, GMV spiked to 151,778 against a baseline of 23,066, a +558.0% surge ($z=29.79$), marking the second consecutive high-severity anomaly day.

WHY

Southeast drove 72.0% of the total move, surging +614.2% vs. baseline, making it the dominant force. South (+472.5%, 13.5% share) and Center-West (+526.7%, 7.2% share) contributed meaningfully alongside Northeast (+294.8%, 6.9% share). All regions moved upward, consistent with a broad promotional event extending from the prior-day spike on 2017-11-23.

RECOMMENDED ACTION

Investigate Southeast's order volume, AOV, and promotional campaign exposure for 2017-11-24 to determine whether this concentration represents genuine demand or fulfillment/reporting inflation before attributing revenue.

■ A spike of similar severity occurred on 2017-11-23 (1 days prior), $z=+5.03$.

▲ 2017-11-25

SPIKE

HIGH SEVERITY

R\$ 60,485

Actual GMV

R\$ 23,214

28-Day Baseline

+160.6%

vs Baseline

+8.35

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+23,060	+72.7%	+125.3%	▲
Northeast	+4,355	+13.7%	+130.9%	▲
South	+3,738	+11.8%	+89.0%	▲
Center-West	+619	+2.0%	+29.4%	▲
North	-75	-0.2%	-9.9%	▼

WHAT HAPPENED

On 2017-11-25, GMV spiked to 60,485 versus a baseline of 23,214 (+160.6%, z=8.35), marking a second consecutive high-severity anomaly following an even larger spike on 2017-11-24 (z=29.79).

WHY

Southeast drove 72.7% of the total move (+125.3% vs baseline), with Northeast contributing 13.7% (+130.9%) and South adding 11.8% (+89.0%). This two-day pattern across all major regions is consistent with a Black Friday demand surge, with North the sole detractor (−9.9% vs baseline, −0.2% share).

RECOMMENDED ACTION

Audit Southeast order volume and seller fulfillment capacity first—given its outsized 72.7% share across both anomaly days, any fulfillment strain or pricing irregularities will concentrate there.

■ A spike of similar severity occurred on 2017-11-24 (1 days prior), z=+29.79.

▲ 2017-11-26

SPIKE

MEDIUM SEVERITY

R\$ 44,852

Actual GMV

R\$ 23,519

28-Day Baseline

+90.7%

vs Baseline

+4.28

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+13,959	+96.3%	+71.5%	▲
South	+1,060	+7.3%	+24.1%	▲
Northeast	-741	-5.1%	-21.0%	▼
Center-West	+640	+4.4%	+29.6%	▲
North	-425	-2.9%	-55.8%	▼

WHAT HAPPENED

On 2017-11-26, GMV spiked to 44,852 against a baseline of 23,519 (+90.7%, z=4.28), marking a second consecutive anomaly day.

WHY

Southeast drove 96.3% of the total move, posting +71.5% above its own baseline, making it the near-singular source of the spike. South (+24.1% vs. baseline) and Center-West (+29.6% vs. baseline) contributed smaller positive shares of 7.3% and 4.4%, respectively. This follows a more severe spike on 2017-11-25 (z=8.35), suggesting sustained elevated demand across this two-day window.

RECOMMENDED ACTION

Immediately audit Southeast's order-level data for 2017-11-25 and 2017-11-26 to identify whether a specific product category, promotional event, or fulfillment anomaly is sustaining the consecutive GMV surge.

■ A spike of similar severity occurred on 2017-11-25 (1 days prior), z=+8.35.

▲ 2017-11-27

SPIKE

MEDIUM SEVERITY

R\$ 47,755

Actual GMV

R\$ 23,963

28-Day Baseline

+99.3%

vs Baseline

+4.21

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+11,592	+70.7%	+56.8%	▲
South	+3,254	+19.8%	+72.4%	▲
Northeast	+1,629	+9.9%	+46.7%	▲
Center-West	-107	-0.7%	-4.9%	▼
North	+35	+0.2%	+4.6%	▲

WHAT HAPPENED

On 2017-11-27, GMV spiked to 47,755 vs. a baseline of 23,963 (+99.3%, z=4.21), marking the second consecutive day of elevated activity following a near-identical spike on 2017-11-26 (z=4.28).

WHY

Southeast drove 70.7% of the total move (+56.8% above its baseline), with South contributing an additional 19.8% (+72.4%) and Northeast 9.9% (+46.7%). Center-West was the sole drag, offsetting -0.7% of the move. This two-day pattern strongly mirrors a sustained Black Friday/Cyber Monday demand surge.

RECOMMENDED ACTION

Investigate Southeast fulfillment and inventory data immediately to confirm supply is keeping pace with the two-day demand surge before stockouts erode conversion.

■ A spike of similar severity occurred on 2017-11-26 (1 days prior), z=+4.28.

▲ 2017-11-28

SPIKE

MEDIUM SEVERITY

R\$ 47,808

Actual GMV

R\$ 24,376

28-Day Baseline

+96.1%

vs Baseline

+3.98

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	+11,047	+71.5%	+52.1%	▲
South	+2,973	+19.2%	+63.2%	▲
Center-West	+1,287	+8.3%	+59.5%	▲
Northeast	+601	+3.9%	+17.2%	▲
North	-451	-2.9%	-57.7%	▼

WHAT HAPPENED

On 2017-11-28, GMV spiked to 47,808 against a baseline of 24,376 (+96.1%, z=3.98), marking the second consecutive day of elevated activity.

WHY

Southeast drove 71.5% of the total move (+52.1% vs baseline), with South contributing an additional 19.2% (+63.2%) and Center-West 8.3% (+59.5%). This follows a similar spike on 2017-11-27 (z=4.21), indicating a multi-day demand surge concentrated in southern and central regions, while North contracted sharply (-57.7% vs baseline), partially offsetting gains.

RECOMMENDED ACTION

Investigate Southeast immediately—confirm whether this two-day surge ties to a specific promotional campaign, seller inventory event, or fulfillment anomaly concentrated in that region.

■ A spike of similar severity occurred on 2017-11-27 (1 days prior), z=+4.21.

▼ 2018-06-30

DROP

MEDIUM SEVERITY

R\$ 13,458

Actual GMV

R\$ 28,083

28-Day Baseline

-52.1%

vs Baseline

-3.82

Robust Z-Score

TOP SEGMENT DRIVERS · decomposed by REGION

Segment	GMV Delta (R\$)	Contribution to Move	Segment % vs Baseline	Dir.
Southeast	-11,556	+73.6%	-60.1%	▼
Northeast	-1,852	+11.8%	-53.0%	▼
Center-West	-1,303	+8.3%	-74.1%	▼
South	-525	+3.3%	-13.1%	▼
North	-456	+2.9%	-72.4%	▼

WHAT HAPPENED

On 2018-06-30, GMV dropped 52.1% below baseline (actual: 13,458 vs. 28,083; z = -3.82), the first anomaly in 214 days.

WHY

Southeast drove 73.6% of the total decline, falling 60.1% below its own baseline, making it the dominant cause. Northeast contributed an additional 11.8% of the move (-53.0%), followed by Center-West at 8.3% (-74.1%). All five regions declined simultaneously, suggesting a platform-wide or date-specific event rather than a localized issue.

RECOMMENDED ACTION

Investigate Southeast transactions on 2018-06-30 first — check for payment gateway failures, promotional expirations, or data pipeline gaps specific to that region on that date.